Antigen

Prof. Dr. Ruqaya M. Al Barzinji
Hawler Medical University
College of Medicine
Department: Microbiology

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Antigens: is a substance that reacts specifically with products of immune response (antibodies and/or T-cell receptors).

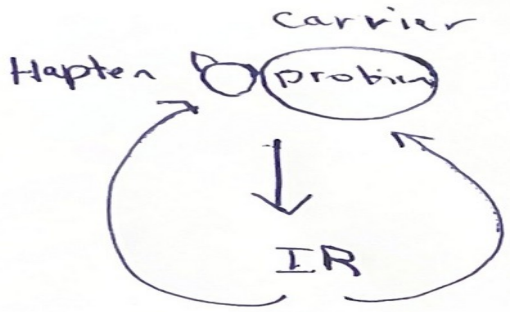
- **Immunogen:** - is defined as any substances that induce a specific immune response mediated by either T-cell (cellular immunity and/or B-cell(Humoral immunity).
- **Haptens**: are antigenic molecules, but incapable by themselves to inducing a specific immune response, they lack immunogenicity.

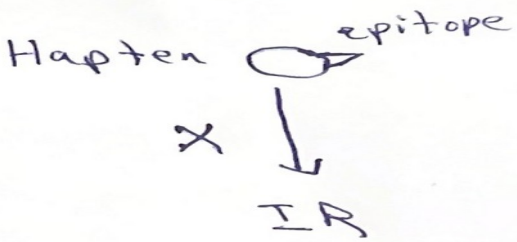
Immunogenicity:

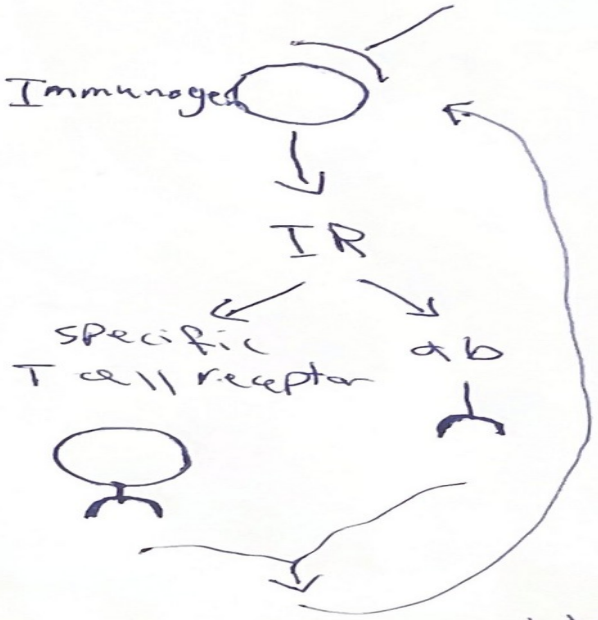
It is the ability to induce a humoral and/or cell mediated immune response.

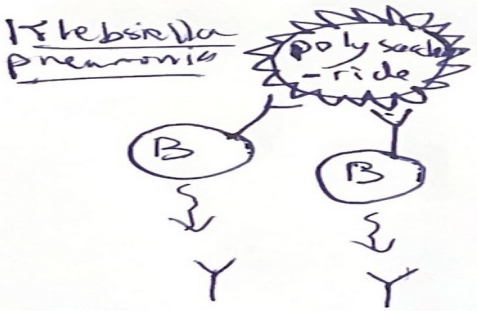
- All immunogens are antigens, but not all antigens are immunogen.

- Some compounds are not able to induce immune response by themselves (hapten), but when they are conjugated with much larger entities, as proteins (carrier) the resultant conjugate induces an immune response that is directed against various parts of the conjugate.
- Many biologically important substances, including drugs, peptide hormones and steroid hormones can function as haptens.













less than 1%



more than 25%

Characteristic features of immunogen:

- **Foreignness**: More foreignness substance more antigenic. The capacity to recognize non-self is accompanied by **tolerance** for self.
- **High molecular weight**: The most active immunogen tend to have a molecular mass above 100,000 Daltons.
- **Chemical complexity**: **Homopolymer** amino acids and sugar are less immunogenic than the **heteropolymers** molecules. Homopolymer lack immunogenicity regardless of their size.
- The **structural complexity** of proteins-primary,

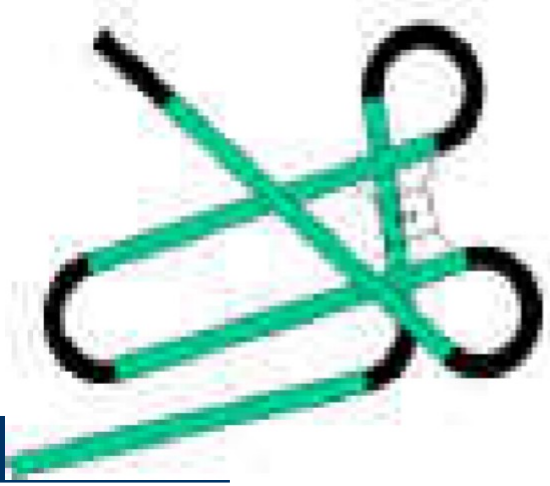
Characteristic features of immunogen:-

- **Dosage and route of administration:**
- **Insufficient** and **high doses** of antigens can not stimulate an immune response or may induce tolerance.
- **Repeated** administration of antigen is required to stimulate a strong immune response.
- **Route of administration:**
- Intravenous, intradermal, subcutaneous, intramuscular and intraperitoneal are mode of administrations.
- Antigen administrated subcutaneously is better than intravenous or intragastric routes.

Characteristic features of immunogen:-

- **The genetic constitution of the host:**
- This factor determines whether a molecule is immunogenic to that host or not.
- The **MHC gene products, genes that encode B-cell and T-cell receptors** and various proteins involved in **immune regulatory mechanisms** and play a central role in determining the degree of responds to an antigen.
- **Susceptibility to antigen processing and presentation**
- Large insoluble macromolecules generally are **more immunogenic** than small and soluble ones because the larger molecules are more phagocytosed and

Epitopes and paratopes:



- Are portions present on the surface of an antigen against which all **immune activity is directed**.
- It also called antigenic determinant area ,determines the **specificity** of antigen-antibody reaction.
- Their size 5-10AA in protein and 3-6 sugar molecules in polysaccharides.
- It is **immunologically active regions** of an immunogen that bind to antigen-specific membrane receptors on **lymphocytes** or to **secreted antibodies**.
- T and B cells recognize different epitopes on the same antigenic molecule
- Exposure to microorganism will generate an immune response to many different epitops and the antiserum will contain different antibodies reactive with each determinant.
- **Paratope**: The part of **the antibody or T-cell receptor**

Classification of antigens:

Antigens can be classified in different ways:-

1 **According to chemical nature:**

- **Proteins:** All proteins are antigenic in appropriate individual.
- **Polysaccharides:** are potentially but not always antigenic.
- **Lipids:** usually are not antigenic but may act as haptens.
- **Nucleic acids:** Are poor immunogens.

2 **On the basis of mode action:**

- **Thymus dependent Ag:** require help from T lymphocytes for induction of immune response.
- **Thymus independent Ag: i**nduce immune response without help of T-cells

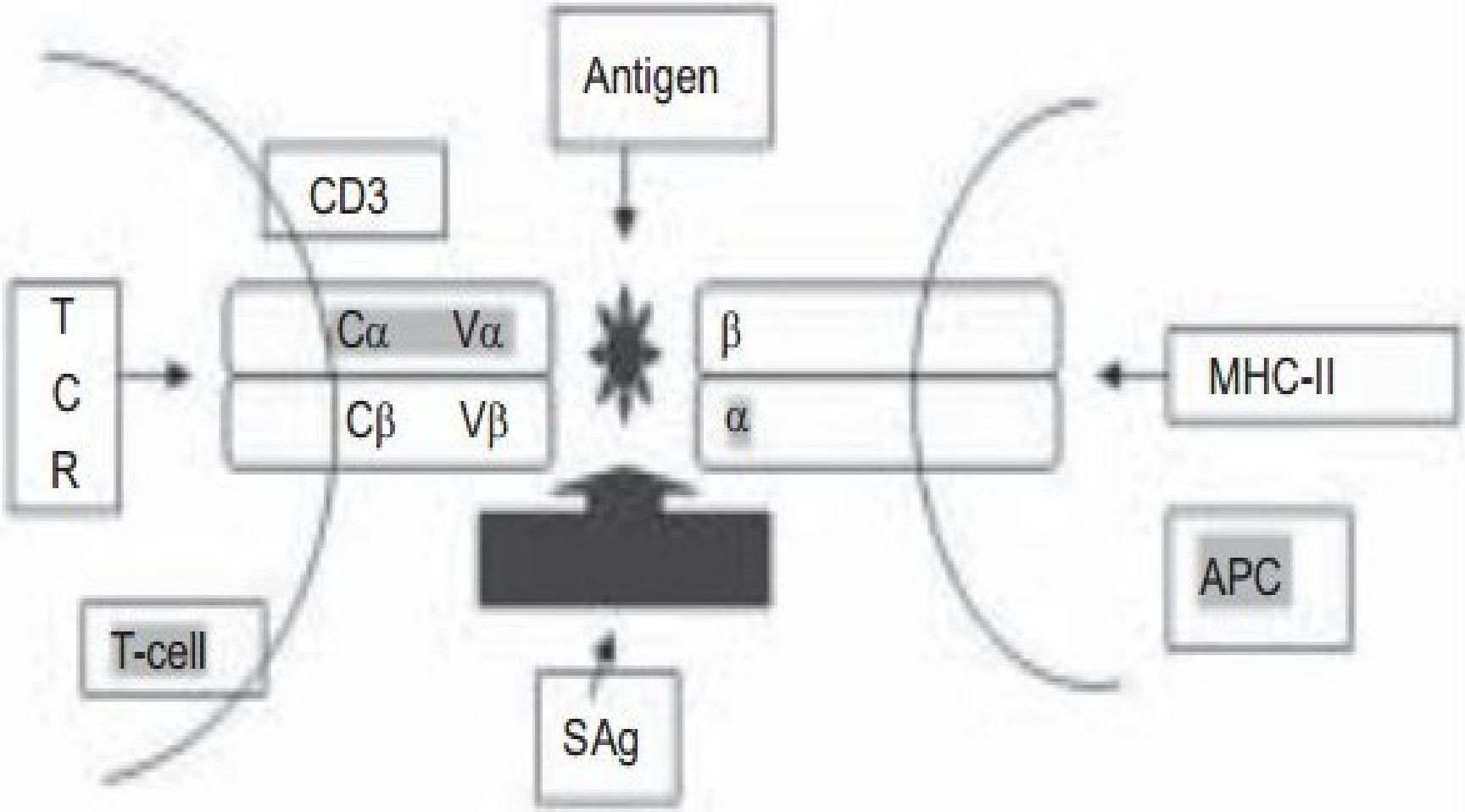
Classification of antigens:

3. On the basis of epitope:

- **Unideterminant univalent:** Single epitope of a given specificity eg. hapten.
- **Unideterminant multivalent:** Two or more determinants of the same kinds eg. polysaccharides.
- **Multideterminant multivalent:** Multiple determinants and different kinds example proteins.

Superantigens:

- Superantigens are microbial proteins of 22-29 Daltons in size an are **potent stimulators of the immune cells** in an unconventional manner by exposure to microorganisms.
- Are antigens that ***polyclonally*** activate a large fraction of the T cells (up to 25%) as staphylococcal enterotoxins (food poisoning and staphylococcal toxic shock toxin (STS).
- These antigens bind directly to class II MHC proteins without internal processing of the toxin and interact with the portions of T cell receptor. This activates T cells to release IL-2 and macrophage to release IL-1 and TNF.
- **A conventional antigen:** Stimulate **a small fraction** (1 in 104 -105) of the **T cell population** (***monoclonal/ oligoclonal)*** response.


Antigen
CD3
T
C
R
Cα
Vα
Cβ
Vβ
β
α
MHC-II
APC
T-cell
SAg

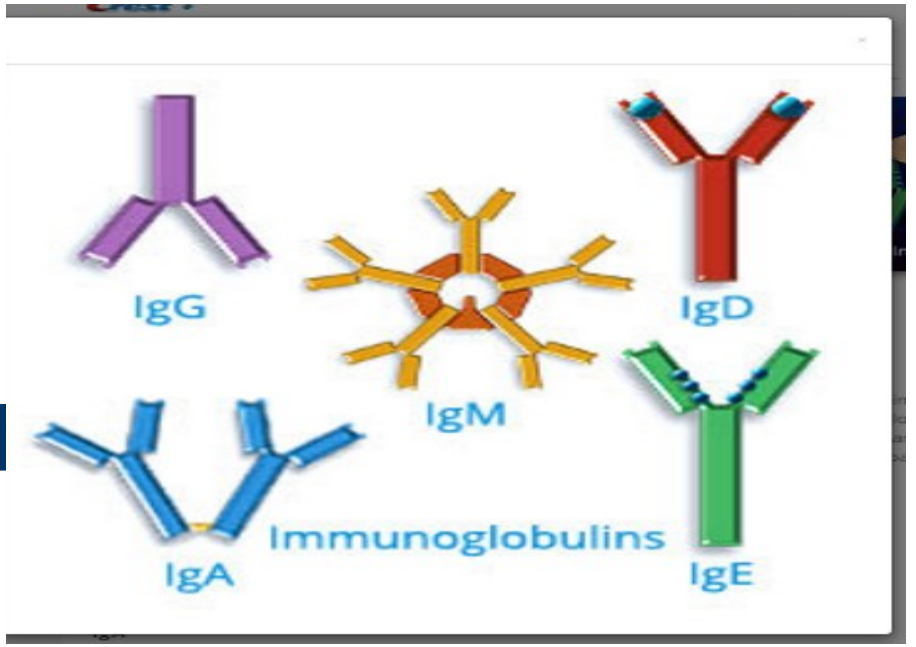
Comparison between conventional and superantigen response

Classical response	Superantigen response
Antigen requires processing by APC	Does not require processing by APC
Antigen recognition and T-cell activation is MHC-II restricted	MHC-II positive cells are required for SAg-induced T-cell activation, but it is not MHC-II restrictive
Small proportion of T-cells become activated (<0.001) and highly regulated response	Massive T-cell activation (20-30% of total T cells) and associated with adverse consequences

Immunoglobulins

It also known as **antibodies**, are glycoprotein molecules produced by plasma cells(WBC).
They act as a critical part of the immune response by specifically recognizing and binding to particular antigens, and aiding in their destruction.
Immunoglobulin occur in two main forms: soluble antibodies and membrane-bound antibodies like BcR(IgM and IgD), it is a heterodimer alpha and beta that enables the cell to transduce the signal and respond to the presence of antigens on the cell surface. The signal generated causes the growth and proliferation of the B cell and antibody production.

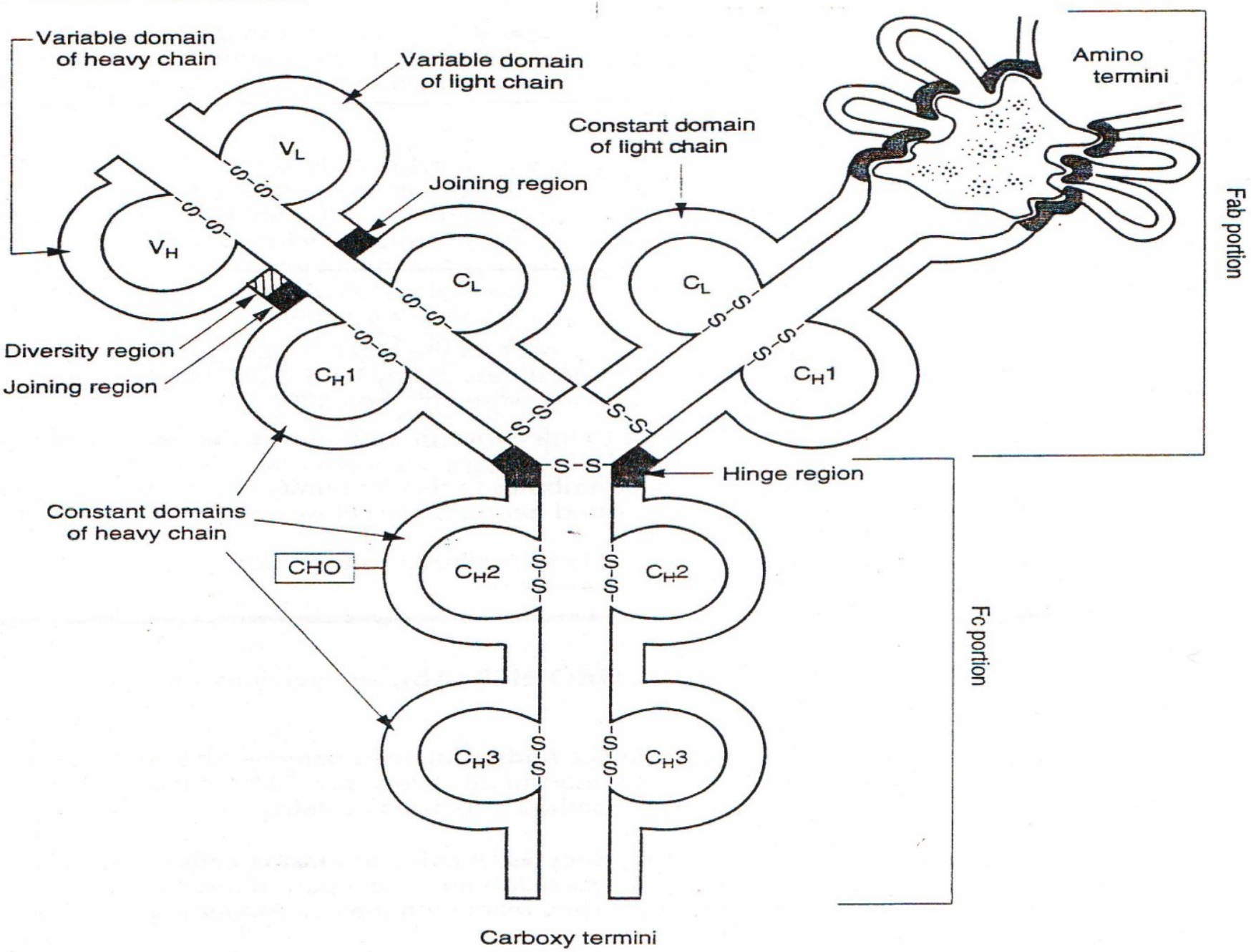
Classes of immunoglobulins



Immunoglobulin G (IgG) [subclass IgG1, IgG2, IgG3, IgG4]
Immunoglobulin A (IgA) [subclass IgA1, IgA2]
Immunoglobulin M (IgM)
Immunoglobulin D (IgD)
Immunoglobulin E (IgE)
Determination of individual classes and subclasses is relevant in assessing primary immunodeficiencies or immune responses.

Basic structure of immunoglobulin:-

- The stem of the Y called the **constant regions**.
- The **variable region** is antigen-binding site make up the tips of the Y's arms.
- Each chain consists of a series domains enclosed by disulfide bonds, each domain consist of 110 AA.
- Each H(heavy) chain has 4 or 5 domains one in the variable region (VH) and 3 or 4 in the constant region (CH1, CH2, CH3, and CH4).
- Each L(light)chain has 2 domains one in the variable region (VL) and one in the constant region (CL).
- The hinge region is the highly flexible portion of the H chain between the CH1 and CH2 domains, IgM and IgE do not possess a hinge.


Variable domain of heavy chain
Variable domain of light chain
Constant domain of light chain
Amino termini
Joining region
Diversity region
Joining region
Hinge region
Constant domains of heavy chain
CHO
Fab portion
Fc portion
Carboxy termini

(Hyde, 2000)

Human Immunoglobulins and their properties.					
	Immunoglobulin	Serum Level (mg/mL)	Molecular Mass (kD)	Activation/Binding of Complement	Half-life (days)
Human	IgA1	0.6–3	160 (monomer)	-	5.5
Human	IgA2	0.06–0.6	160 (monomer)	-	5.5
Human	IgM	1.5	970 (pentamer)	+++	5–10
Human	IgE	5×10^{-5}	188	-	2
Human	IgG1	3.8–11.4	146	++	23
Human	IgG2	1.5–6.9	146	+	23
Human	IgG3	0.2–1.7	165	+++	7
Human	IgG4	0.08–1.4	146	-	23

+++ = Very strong affinity; ++ = Strong affinity; + = Moderate affinity; - = No affinity

Immunoglobulin classes and subclasses:-

On the bases of AA sequences at **constant region**:

There are 2 types of **light chain kappa κ and lambda λ**, while **heavy chains** are of **5 types**

Gamma γ, Alpha α, Meu μ, Delta δ and Epsilon ε.

So there are 5 classes or isotypes of Igs are IgG, IgA, IgM, IgD and IgE.

Igs G, D, and E are monomer structure. IgG has 4 subclasses IgG1, IgG2, IgG3 and IgG4.

Isotype class switching

Some B cells can undergo a “class switch” that causes the expression of a new antibody isotype. For example, the antibody isotype could switch from an IgM to an antibody of all possible classes (e.g., IgG1,...IgG4, IgE).
During this switch, the constant region of the heavy chain is changed, but not the variable region of the heavy chain. This switch does not affect the antibody’s specificity for its antigen, but it does alter the effector functions that each class of antibody can execute.

Isotype class switching

The antibody class switch is critically dependent on the type of cytokine that is present.

Various cytokines, such as IL-4, IL-5, IFN-gamma and TGF-beta, are known to be responsible for class switching.

IgG and IgM

IgG: It is a monomer and a major Igs in the blood, it **crosses the placenta** in the fetus.
Play role in opsonization and involved in Antibody-dependent cell-mediated cytotoxicity (ADCC).
Activation of complement , neutralization of toxin and antiviral activity.
IgM: It is a pentamer and the first Ig produced in the immune response. Play role in agglutination, activation of complement and neutralization of toxins.

IgA: IgA exists in two isotypes, IgA1 and IgA2.

IgA1 is monomeric and predominates in serum
IgA2 is dimeric and a major Ig in external secretions such as saliva, sweat, gastric fluid, tears, nasal fluids, fluids of the lung and colostrum, genito-urinary tract, and gastro-intestinal tract to provides localized antibody protection on mucosal surfaces, by preventing the microorganism from **adhering to, and penetrating**, the mucosal epithelial lining.
. The dimer jointed together by a J-chain and contains a secretary component or S piece by which is secreted from the epithelial cells.

Secrotary IgA biological functions:-

Bacteriocidal activity: It has bacteriocidal activity against G-ve B.

Antiviral activity:It prevent viruses from entering into the host cells.

Agglutination: Secretary IgA is also an efficient agglutinating Ab.

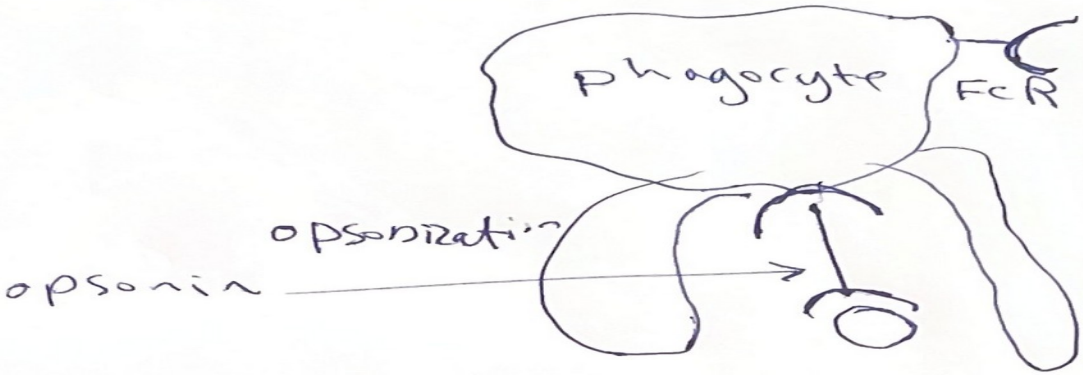
Complement activation: Activate the alternative pathway of the complement, it is present in colostrums and milk to protect the neonate.

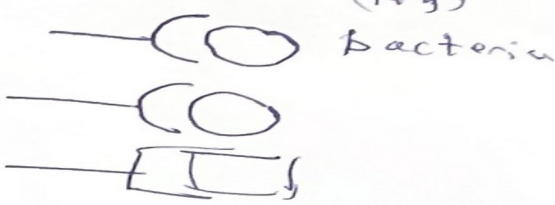
Serum IgA: In the blood, IgA interacts with an Fc receptors called **FcαRl (CD89)**, that expressed on immune effector cells, to initiate inflammatory reactions.

Ligation of FcαRI by IgA containing immune complexes causes antibody dependent cell mediated cytotoxicity (ADCC).

IgD and IgE

IgD: Present in the serum in low and variable concentration.
It is a major surface Ig of immature B-cells and involved in their differentiation.
IgD biological function in the serum is not known.
IgE: Known as reagenic antibody.
It is the **lowest** among all immunoglobulins.
It has a unique ability of the Fc portion of IgE to bind with very **high affinity** to mast cells and basophils.
It protects against invasion by parasites as helminthes.
It plays a role in type I hypersensitivity.



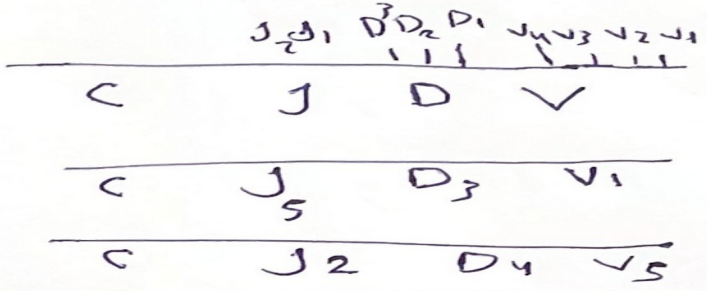




(ADCC)



Igs diversity



Immunoglobulin genes

	IgG	IgA	IgM	IgD	IgE
H-chain	γ	α	μ	δ	ϵ
H-chain subclasses	$\gamma_1, \gamma_2, \gamma_3, \gamma_4$	α_1, α_2	—	—	—
H-chain allotypes	Gm	Am	—	—	—
J chain	—	+	+	—	—
Carbohydrate (%)	3	7	12	13	11
Serum concentration (mg/dl)	1200	200	120	3	0.05
Serum half-life (days)	21†	6	10	3	2

H chain = heavy chain; J chain = joining chain.

†IgG3 has a half-life of 7 days.

(Hyde, 2000)

Immunoglobulin diversity

- All microbes can trigger an antibody response.
- Recognition and eradication of many different types of microbes require diversity among antibodies.
- Human generate about 10 billion different antibodies, each capable of binding a distinct epitope of an antigen.
- Diverse pool of antibodies from a relatively small number of antibody genes achieved by **V(D)J recombination.**